PART 2 · VISUAL EXPLAINER

How curves work, and how Track It moves things along them

A picture-first walk through what Blender stores inside a curve, how Follow Path turns it into motion, and exactly what each Track It feature adds on top. Every later Part builds on this one.

2.1 What a curve object contains
2.2 How a Bézier segment is shaped
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2.1 What a curve object contains

A “curve” in Blender is three nested things. The **object** gives it a place in the world, the **curve data** holds the settings, and inside that is a list of **splines**, each a chain of points. Track It only cares about the first spline.



Figure 2.1 The object → data → spline → point nesting. Tested in session 4: sweeping offset_factor 0→1 on a two-spline curve only ever travelled spline 0.

What each Bézier point stores

- `co` : the point's position (in the curve's *local* space)
- `handle_left` , `handle_right` : steer the shape either side
- `tilt` : roll around the track direction. This is your **superelevation**
- `radius` : scales bevel, and scales objects if “Curve Radius” is on (Track It leaves it off)

Bézier vs Poly/NURBS (“paths”)

A **Path** object (Add → Curve → Path) is a NURBS spline. It starts straight, which is why the switching tests used them. Its points have no handles; the shape comes from point weights. The add-on handles both: Bézier gets the cubic formula, the others are treated as straight lines between points, which is only an approximation for NURBS (Part 4).

2.2 How a Bézier segment is shaped

Between two neighbouring points, Blender draws a **cubic Bézier**. It starts at point A heading toward A's right handle, ends at point B arriving from B's left handle, and blends smoothly in between. A single number **t** runs from 0 at A to 1 at B:

```
position(t) = (1−t)³·A  +  3(1−t)²t·A.handle_right  +  3(1−t)t²·B.handle_left  +  t³·B
```

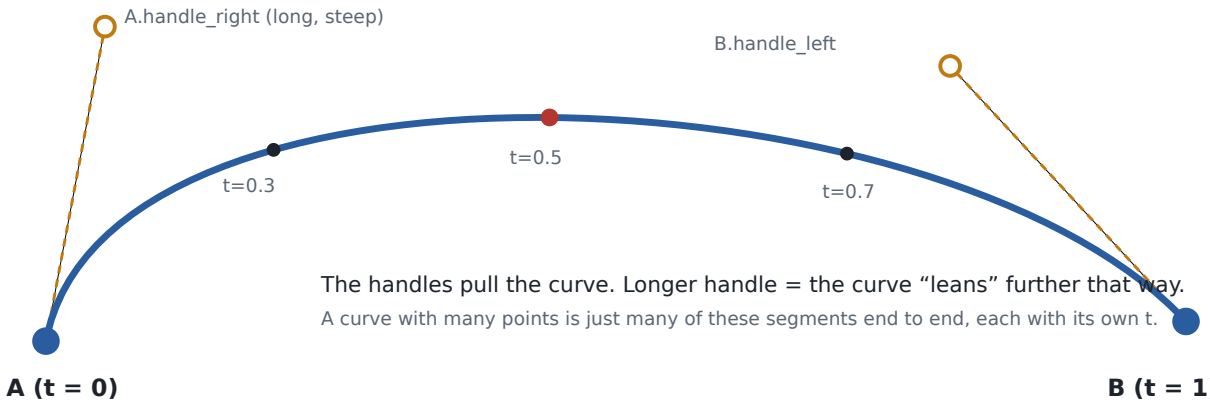


Figure 2.2 One Bézier segment drawn from its four controls, which is exactly what `build_arc_table` evaluates in the add-on.

2.3 Parameter t vs real distance

This is the single most important idea for the whole system. **Equal steps of t are not equal steps of distance.** Where handles are long or points are bunched, t races ahead; elsewhere it crawls. In the segment above, **t = 0.5 is only 46% of the way along**.

Top: 10 equal steps of t Bottom: 10 equal steps of distance (arc length)

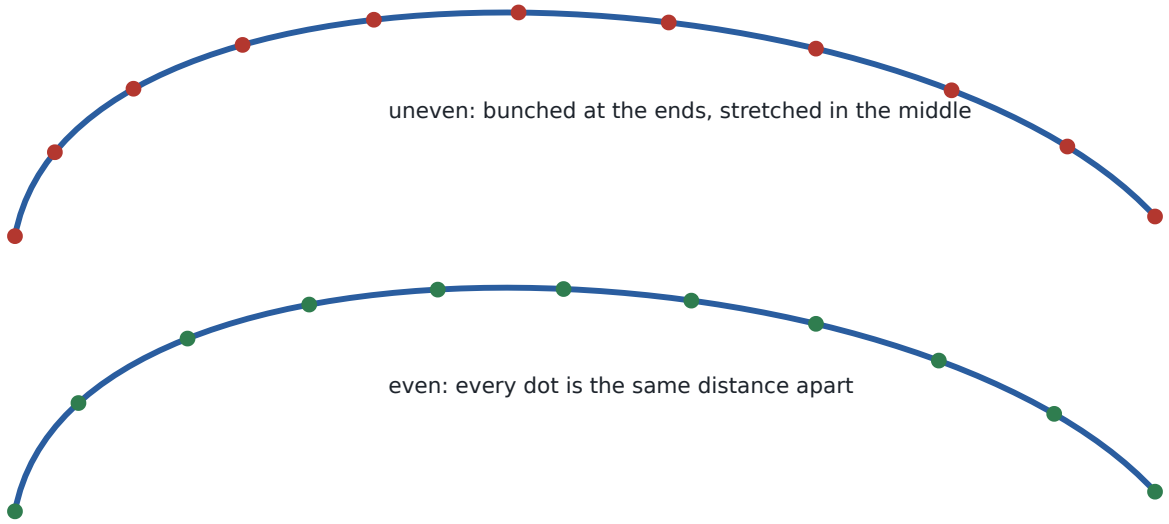


Figure 2.3 Real geometry, computed. A train driven by raw t would speed up and slow down on its own. Follow Path's `offset_factor` uses the green spacing, and so does the add-on's arc table.

CONFIRMED IN THE PROJECT

offset_factor = distance along the curve ÷ total curve length. Setting 0.25 puts the object a quarter of the way along in metres, whatever the handles look like. That is why an offset in metres converts to a factor with one division, `metres / length`, which is exactly what the pivot and consist code does.

2.4 What Blender actually follows

Blender doesn't evaluate the smooth formula at render time. It chops each segment into **`resolution_u`** straight pieces and follows that **polyline**. With a low resolution, a curve's real “followed” shape differs slightly from the smooth one:

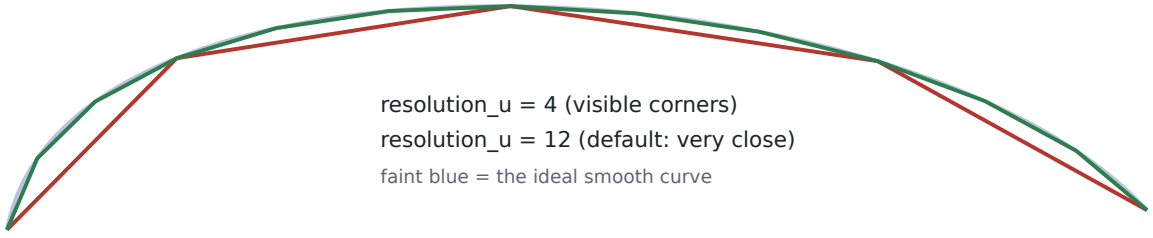


Figure 2.4 Keep track curves at resolution 12 or higher. Proposal (Part 4): build the arc table from the *evaluated* polyline, the same data Follow Path uses, so the add-on and Blender agree exactly.

2.5 Follow Path, the engine

Everything Track It does comes down to setting properties on Blender's own **Follow Path** constraint. The constraint has two modes, and the project chose the second one:

Mode A: “Animate Path” (default)

Position comes from the timeline:

`frame − offset → eval_time / path_duration`

frame 1 — frame 100

✗ offset is measured in **frames**
✗ object always starts at the curve start
✗ moving depends on the playhead

(the “−99.8 offset” Cube.007 was in this mode)

Mode B: Fixed Position ✓ (Track It)

use_fixed_location = True

`offset_factor 0.0 … 1.0 = where it sits`

0.0 — 0.5 — 1.0

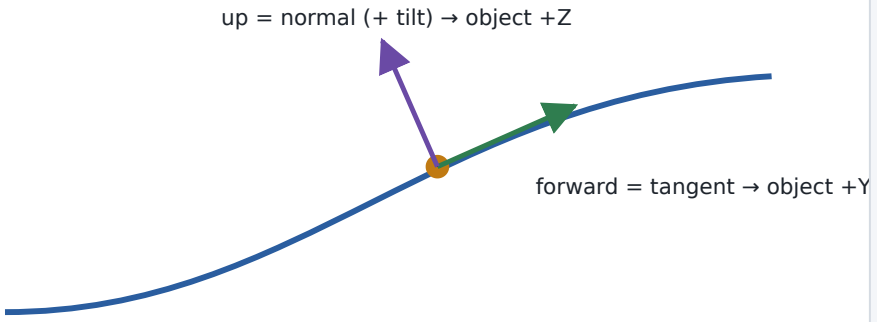
✓ a plain number you can keyframe
✓ can be set to “where it is now”
✓ frame-independent; easing = speed

forward_axis FORWARD_Y · up_axis UP_Z · Follow Curve on

Figure 2.5 The two Follow Path modes. Every constraint Track It creates uses Fixed Position.

2.6 Orientation, tilt and banking

With **Follow Curve** on, the constraint also rotates the object. At the object's point on the curve Blender builds a little coordinate frame: **forward** = the curve's direction there (the tangent), **up** = the curve's normal, then rolled by the point's **tilt**. The object's local **+Y** is aimed forward and its **+Z** up.



Side/top view: the frame travels with the object as offset_factor changes.

Looking down the track



Tilt set on curve points in edit mode (Ctrl+T) = superelevation. It banks the train for free, no bank plane needed.

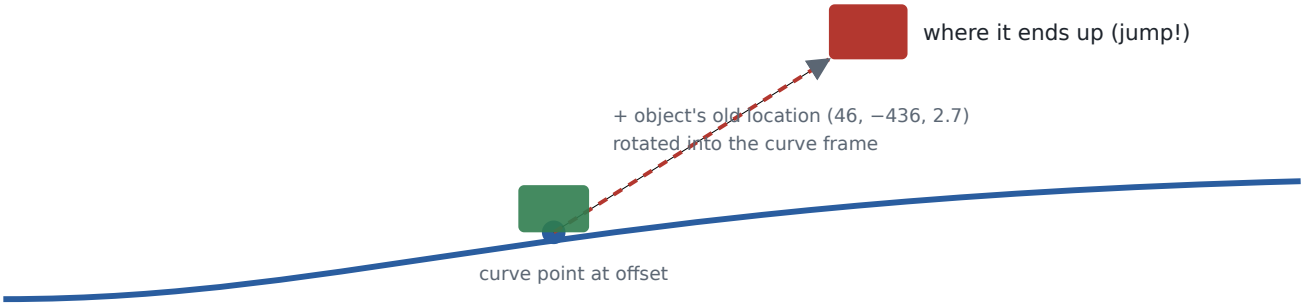
Figure 2.6 This replaced the legacy direction plane and bank plane. The curve itself now carries heading and banking.

WATCH OUT

On curves that climb or twist sharply, the “up” direction can flip or wander (Blender's `twist_method`: Minimum / Tangent / Z-Up). For railways, **Z-Up** is usually the calm choice, since track is mostly flat with gentle grades. Track Doctor (Part 4) should warn when it isn't set.

2.7 Why objects “jump” on attach

The early sessions fought this for a whole day (the “error 1–4” cones). The reason: with Follow Path active, the object's **own** location and rotation don't disappear. They are applied **on top of** the curve point, inside the curve's moving frame.



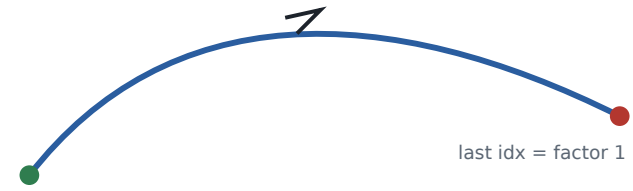
Final position = curve point · curve frame · (object's own local transform)

Fix used by Attach: compute the offset first, then set the object's local location and rotation to zero, so the constraint alone places it.

Figure 2.7 The same rule explains the “re-attach sits wrong” bug (baked rotation left on the object) and the pivot re-attach jump. Both were fixed by resetting to identity every time.

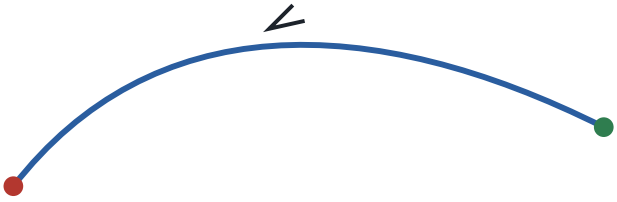
2.8 Direction, point order and extra splines

Point order = direction



Extruding back from the start inserts the new points at index 0.

Switch Direction flips everything



factor f becomes 1 − f. Keyframed offsets would then run backwards. Track Doctor should warn before direction is changed on a used track.

Two practical consequences: a curve's “forward” is set by how it was drawn, and joining two curves (Ctrl+J / F) can leave them facing opposite ways. The Track Network (Part 4) stores, for each connection, whether factor increases or decreases across it. Then direction never has to match.

2.9 Track It's Attach, step by step

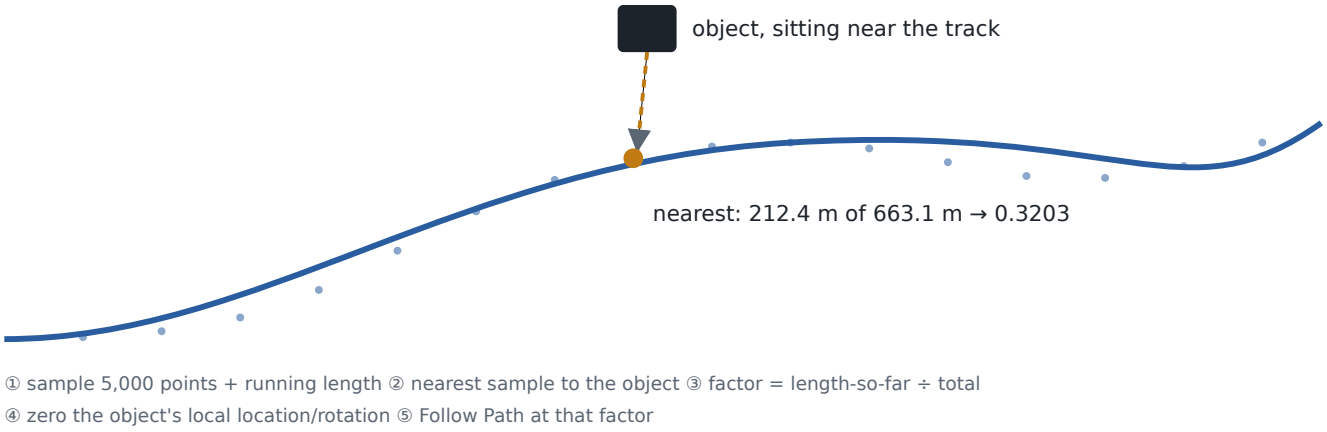


Figure 2.8 What `OT_AttachToTrack_Action` does (numbers illustrative). The same nearest-point math powers Nearby Curves, held-offset sync and Mouse Drag.

Attach also reuses an existing constraint for that curve if the object has been attached before, and sets every other Follow Path to influence 0. That is covered in §2.11.

2.10 Animating: offset_factor over time

Once attached, animating a train means keyframing `offset_factor`. The F-curve in the Graph Editor then reads directly as **position over time**, and its slope is speed:

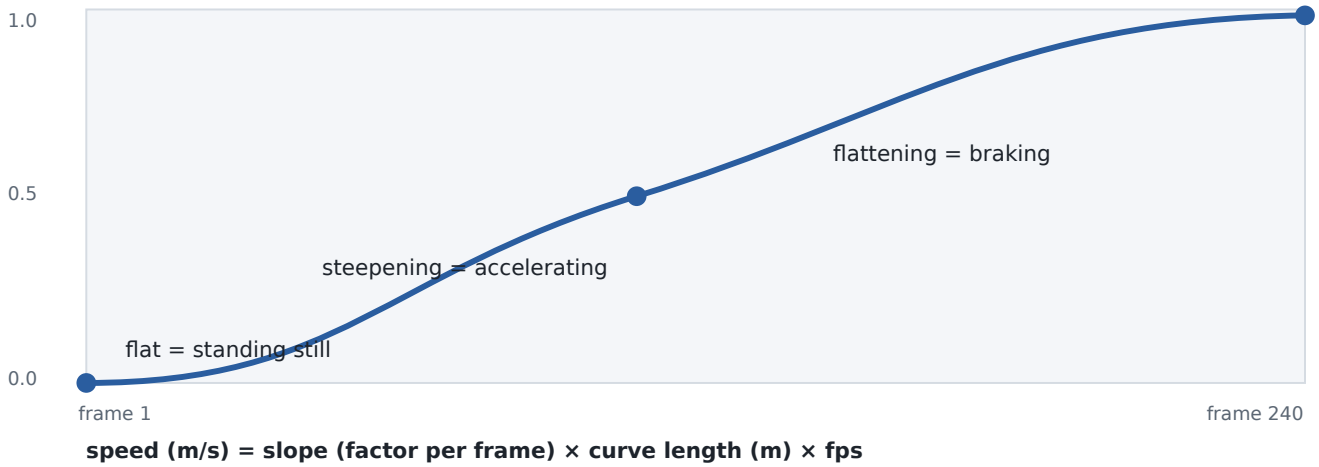


Figure 2.9 One F-curve per held constraint. The catch: the same slope means a different speed on a 100 m curve than on a 2 km one. That is why Part 7 proposes animating **route distance in metres** instead.

The **Drag Slider** is a live way to change that value without keyframing: every slider update nudges offset_factor by `(slider − 0.5) × 2 × Drag Power × Δt`. Nothing is written to the timeline until you press Keyframe.

2.11 Many constraints, one active: how switching works

An object can carry several Follow Path constraints at once, one per curve it has been near. Blender blends constraints by **influence**. Track It keeps exactly one at 1.0 and the rest at 0.0, and switches by swapping which one is 1.0:

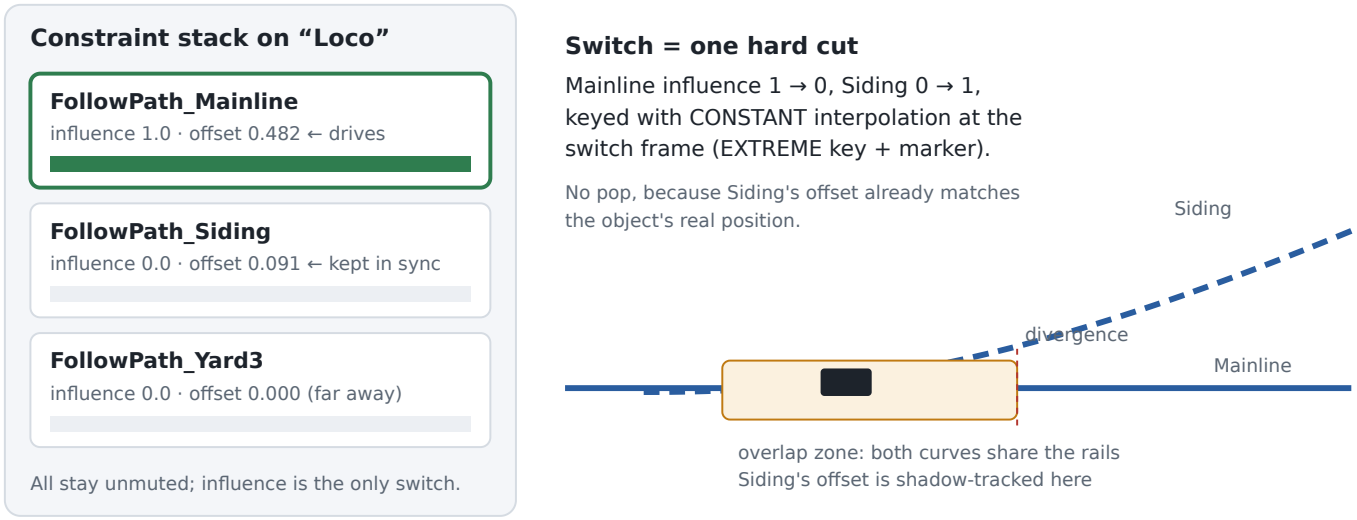


Figure 2.10 Influence switching. Held curves have their offset recomputed every tick from the object's world position (“shadow tracking”), so whichever one is switched to is already in the right place.

2.12 Pivots and wagons: derived points

A pivot (a bogie, pony truck and so on) isn't dragged by the user. The add-on **derives** it from the root: “the same curve, N metres ahead or behind.” Because offset_factor is distance ÷ length, that is one line of arithmetic, and the pivot gets its *own* Follow Path so it takes the curve's heading and tilt at *its* spot:

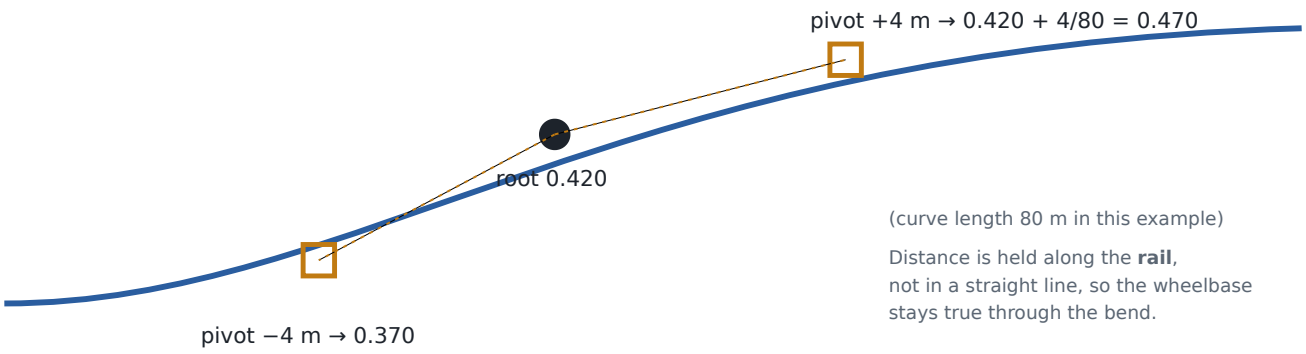
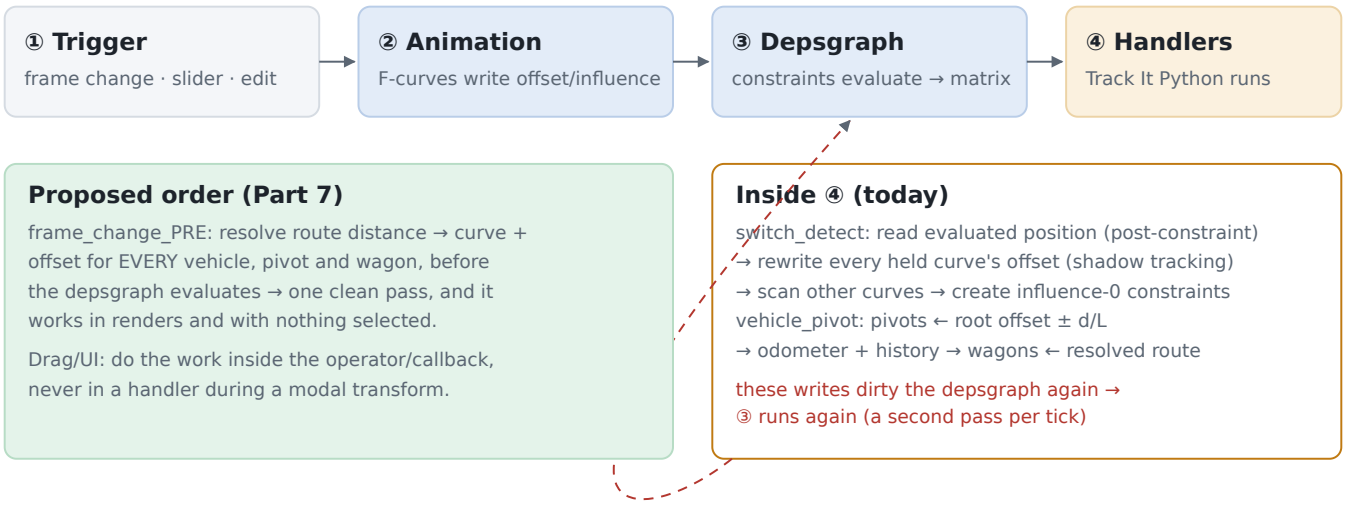


Figure 2.11 `_track_it_sync_vehicle_pivots`. When the result falls past 0 or 1 it is clamped to the end of the root's curve today. Wagons avoid that by looking up the **route history** (Parts 5 and 8).

2.13 One frame, start to finish

When the playhead moves, or you drag the slider, Blender and Track It take turns. Knowing this order explains most “why is it one frame behind” puzzles:



The “one tick stale” bug (matrix_world lagging a fresh offset write) is this ordering. The slider code forces view_layer.update() to get around it.

Figure 2.12 Evaluation order today vs proposed.

2.14 Rules of thumb for track curves

Rule	Why
One spline per curve object	Follow Path reads only `splines[0]`. A branch is a separate object.
Bézier, resolution_u ≥ 12	Blender follows the polyline, so low resolution gives corners (Fig. 2.4).
Twist method Z-Up, banking by tilt	Stable “up”; superelevation is explicit and keyframe-free.
Apply scale on track objects	Non-uniform object scale distorts lengths and tilt. Keep scale (1,1,1).
Real switches need real overlap	The overlap span must be at least as long as the vehicle's longest pivot offset, and longer for trains (Part 5).
Plain joints: endpoints touching	End-to-end pieces with only one way on can be short and modular.
Don't flip direction on used track	Keyframed factors would run backwards (Fig. 2.8).
Keep “Curve Radius” off on constraints	Otherwise the point radius scales the train.

SUMMARY IN ONE SENTENCE

A track is a single spline. Follow Path turns a **0–1 distance fraction** into a position and an orientation. Track It's whole job is to **choose that fraction** (Attach, Drag), **choose which curve's fraction counts** (influence switching), and **derive the fractions of everything that rides along** (pivots, wagons).